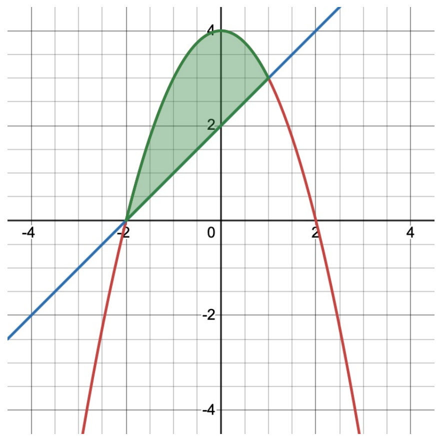
Math 252, First Examination Review

Directions: Respond to each question in the space provided, under each question. You may use the back of the page if necessary. All calculations and mathematical work must be shown and justified, including citation of well known Theorems or properties that have been used for solving the problem. Only a TI-30X (type) scientific calculator may be used on this exam. This is a closed note, closed book exam. No notes, laptops, smartwatches, mobile devices, or any other unapproved electronic device may be used while taking this exam.

1. Consider the graphs of of $y = 4 - x^2$ and $y = x + 2$.

   (a) Sketch a graph of the functions and determine x-values at which the graphs intersect.
   Solution: The graph is as follows,



   The points of intersection satisfy

$$\begin{aligned} 4 - x^2 &= x + 2 \\ -x^2 - x + 2 &= 0 \\ x^2 + x - 2 &= 0 \\ (x+2)(x-1) &= 0. \end{aligned}$$

   Therefore, the graphs intersect at

$$x = -2, \qquad x = 1.$$

   (b) Let R be the region bounded between the two graphs. Write **AND** evaluate an integral for the area of R.
   Solution: From the graph, we see that $4 - x^2 \geq x + 2$ on $[-2, 1]$. Thus the area of the region bounded between the curves is

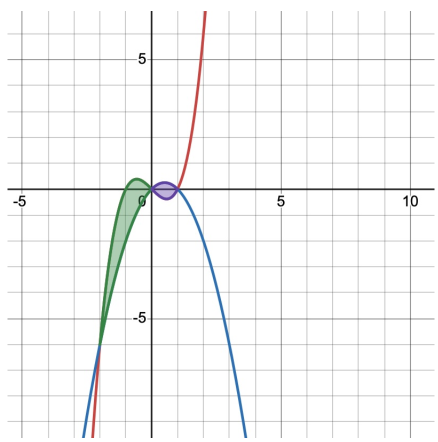
$$\begin{aligned} A &= \int_{-2}^{1} \left[(4 - x^2) - (x + 2)\right] \, dx \\ &= \int_{-2}^{1} \left(2 - x - x^2\right) \, dx. \end{aligned}$$

Evaluating,

$$\begin{aligned}
A &= \left[2x - \frac{x^2}{2} - \frac{x^3}{3}\right]_{-2}^{1} \\
&= \left(2 - \frac{1}{2} - \frac{1}{3}\right) - \left(-4 - \frac{4}{2} + \frac{8}{3}\right) \\
&= \frac{7}{6} - \left(-\frac{10}{3}\right) \\
&= \frac{7}{6} + \frac{20}{6} \\
&= \frac{27}{6} \\
&= \frac{9}{2}.
\end{aligned}$$

2. Let $f(x) = x^3 - x$ and $g(x) = x - x^2$.

   (a) Sketch a graph of the functions and determine x-values at which the graphs intersect.
   Solution:



The points of intersection satisfy

$$\begin{aligned}
x^3 - x &= x - x^2 \\
x^3 + x^2 - 2x &= 0 \\
x(x^2 + x - 2) &= 0 \\
x(x+2)(x-1) &= 0.
\end{aligned}$$

Therefore, the graphs intersect at

$$x = -2, \qquad x = 0, \qquad x = 1.$$

   (b) Let R be the region bounded between the two graphs. Write **AND** evaluate an integral for the area of R.
   Solution:

From the graph, we see that $f(x) \geq g(x)$ on $[-2, 0]$ and $g(x) \geq f(x)$ on $[0, 1]$. Therefore, the area of R is

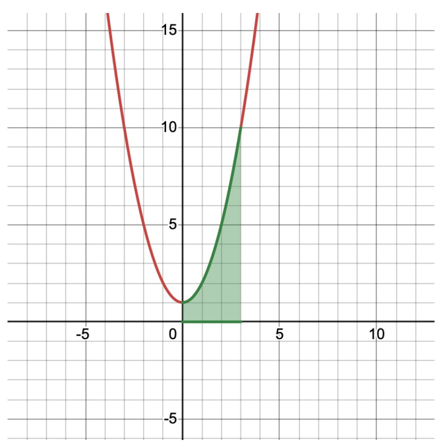
$$\begin{aligned} A &= \int_{-2}^{0} \big(f(x) - g(x)\big)\, dx + \int_{0}^{1} \big(g(x) - f(x)\big)\, dx \\ &= \int_{-2}^{0} \left[(x^3 - x) - (x - x^2)\right]\, dx + \int_{0}^{1} \left[(x - x^2) - (x^3 - x)\right]\, dx \\ &= \int_{-2}^{0} \left(x^3 + x^2 - 2x\right)\, dx + \int_{0}^{1} \left(2x - x^2 - x^3\right)\, dx. \end{aligned}$$

Now evaluate each integral:

$$\begin{aligned} A &= \left[\frac{x^4}{4} + \frac{x^3}{3} - x^2\right]_{-2}^{0} + \left[x^2 - \frac{x^3}{3} - \frac{x^4}{4}\right]_{0}^{1} \\ &= \left(0 - \left(4 - \frac{8}{3} - 4\right)\right) + \left(1 - \frac{1}{3} - \frac{1}{4}\right) \\ &= \frac{8}{3} + \frac{5}{12} \\ &= \frac{32}{12} + \frac{5}{12} \\ &= \frac{37}{12}. \end{aligned}$$

3. Let D be the region bounded between the graph of $y = x^2 + 1$ for $0 \leq x \leq 3$.

   (a) Sketch a graph of the region clearly indicating the region D, and set up an integral for the volume of the solid obtained by revolving D around the line $y = 10$. DO NOT EVALUATE THE INTEGRAL.
   Solution:



Revolving D around the line $y = 10$, we use washers.
The outer radius is

$$R(x) = 10$$

and the inner radius is

$$r(x) = 10 - (x^2 + 1) = 9 - x^2.$$

Therefore, the volume is

$$V = \pi \int_0^3 \left[10^2 - (9 - x^2)^2\right] \, dx.$$

(b) Set up an integral for the volume of the solid obtained by revolving D around the line $x = -1$. DO NOT EVALUATE THE INTEGRAL.
Solution: Revolving D around the line $x = -1$, we use cylindrical shells.
The radius of a shell is

$$r(x) = x + 1$$

and the height of a shell is

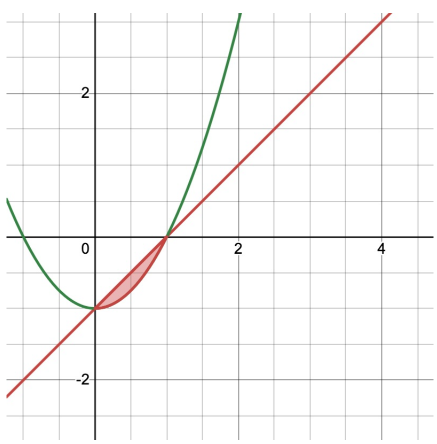
$$h(x) = x^2 + 1.$$

Therefore, the volume is

$$V = 2\pi \int_0^3 (x + 1)(x^2 + 1) \, dx.$$

4. Let R be the region bounded between the graphs of $f(x) = x - 1$ and $g(x) = x^2 - 1$.

(a) Sketch a graph of the region clearly indicating the region R, and set up an integral for the volume of the solid obtained by revolving R around the line $y = -2$. DO NOT EVALUATE THE INTEGRAL.
Solution:



First, the graphs intersect when

$$\begin{aligned} x - 1 &= x^2 - 1 \\ x &= x^2 \\ x^2 - x &= 0 \\ x(x - 1) &= 0. \end{aligned}$$

Therefore, the graphs intersect at

$$x = 0, \qquad x = 1.$$

On the interval $[0, 1]$, we have $f(x) \geq g(x)$. Revolving R around the line $y = -2$, we use washers.

The outer radius is

$$R(x) = (x - 1) - (-2) = x + 1,$$

and the inner radius is

$$r(x) = (x^2 - 1) - (-2) = x^2 + 1.$$

Therefore, the volume is

$$V = \pi \int_0^1 \left[(x + 1)^2 - (x^2 + 1)^2\right] \, dx.$$

(b) Set up an integral for the volume of the solid obtained by revolving R around the line $x = 2$.

Solution: Revolving R around the line $x = 2$, we use cylindrical shells.

The radius is

$$r(x) = 2 - x,$$

and the height is

$$h(x) = (x - 1) - (x^2 - 1) = x - x^2.$$

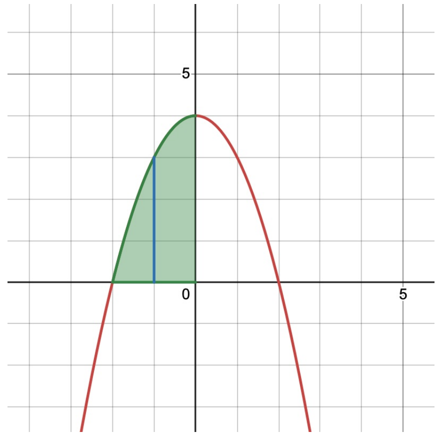
Therefore, the volume is

$$V = 2\pi \int_0^1 (2 - x)(x - x^2) \, dx.$$

5. The base of a solid is bounded between the positive y-axis, the graph of $y = 4 - x^2$, and the negative x-axis for $-2 \leq x \leq 0$, and whose cross-sections are squares with a side that runs perpendicular to the x-axis.

(a) Sketch a graph of the base, and another picture of the cross-section.

Solution:



(b) Set up an integral for the volume of the solid . DO NOT EVALUATE IT. Note: $A_{square} = s^2$.
Solution: Since the cross-sections are perpendicular to the x-axis, we use vertical slices. For each x in the interval $[-2, 0]$, the side length of the square is the distance from the negative x-axis to the graph of $y = 4 - x^2$.
Thus,

$$s(x) = 4 - x^2.$$

Since the cross-sections are squares, the cross-sectional area is

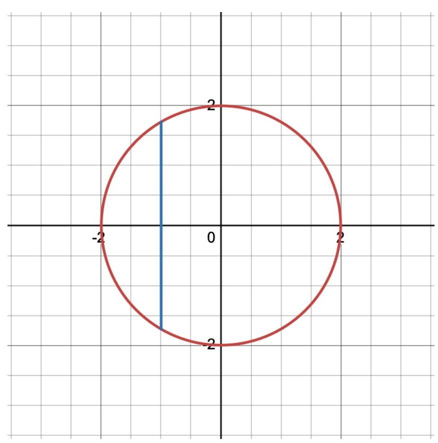
$$A(x) = s(x)^2 = \left(4 - x^2\right)^2.$$

Therefore, the volume of the solid is

$$V = \int_{-2}^{0} \left(4 - x^2\right)^2 \, dx.$$

6. The base of a solid is the disk of radius 2 (i.e. $x^2 + y^2 = 4$).

(a) Sketch a graph of the base, and another picture of the cross-section.
Solution:



(b) If the cross-sections are isoscelese triangles, perpendicular to the x-axis, write an integral for the volume of the solid. DO NOT evaluate the integral. ($A_{eq.\,triangle} = \frac{1}{2}s^2$)

Solution: Since the cross-sections are perpendicular to the y-axis, we use horizontal slices. From

$$x^2 + y^2 = 4,$$

we solve for x:

$$x = \pm\sqrt{4 - x^2}.$$

Thus, the side length of the triangle is

$$s(x) = \sqrt{4 - x^2} - \left(-\sqrt{4 - x^2}\right) = 2\sqrt{4 - x^2}.$$

Since the cross-sections are isosceles triangles with area

$$A = \frac{1}{2}s^2,$$

we have

$$A(y) = \frac{1}{2}\left(2\sqrt{4 - x^2}\right)^2.$$

Therefore, the volume is

$$V = \int_{-2}^{2} \frac{1}{2}\left(2\sqrt{4 - x^2}\right)^2\, dx.$$

7. We have looked at a lot of Riemann Sums during this unit. Consider the following Riemann Sum, $\displaystyle\sum_{i=1}^{n}\left(\frac{2i}{n}\sqrt{1 + \frac{2i}{n}}\right)\cdot\frac{2}{n}$, where $0 \le x \le 2$.

(a) What is the Δx and and determine if left end point rule, right endpoint rule, or midpoint rule was used.

Solution: Since the interval is $0 \le x \le 2$, we have

$$\Delta x = \frac{2 - 0}{n} = \frac{2}{n}.$$

Also,

$$x_i = 0 + i\Delta x = \frac{2i}{n}.$$

Since the sum uses $x_i = \frac{2i}{n}$, this is a right endpoint Riemann Sum.

(b) Complete the following statement: The Riemann Sum will converge to the integral ________________, when _______________ (DO NOT EVALUATE THE INTEGRAL OR THE SUM).
Solution: The function is

$$f(x) = x\sqrt{1+x}.$$

Therefore, the Riemann Sum will converge to the integral

$$\lim_{n\to\infty} \sum_{i=1}^{n} \left(\frac{2i}{n} \sqrt{1 + \frac{2i}{n}} \right) \cdot \frac{2}{n} = \int_0^2 x\sqrt{1+x}\, dx$$

8. We have looked at a lot of Riemann Sums during this unit. Consider the following Riemann Sum, $\displaystyle\sum_{i=1}^{n} \frac{(2k-1)\pi}{2n} \sin^2\left(\frac{(2k-1)\pi}{2n}\right) \cdot \frac{\pi}{n}$, where $0 \le x \le \pi$.

(a) What is the Δx and determine if left end point rule, right endpoint rule, or midpoint rule was used?
Solution:
We are given the Riemann Sum

$$\sum_{k=1}^{n} \frac{(2k-1)\pi}{2n} \sin^2\left(\frac{(2k-1)\pi}{2n}\right) \cdot \frac{\pi}{n}.$$

Since the interval is $0 \le x \le \pi$, we have

$$\Delta x = \frac{\pi - 0}{n} = \frac{\pi}{n}.$$

The sample points are

$$x_k^* = \frac{(2k-1)\pi}{2n} = \left(k - \frac{1}{2}\right) \Delta x.$$

Therefore, the midpoint rule was used.

(b) Complete the following statement: The Riemann Sum will converge to the integral ________________, when _______________ (DO NOT EVALUATE THE INTEGRAL OR THE SUM).
Solution: The function is

$$f(x) = x\sin^2(x).$$

Thus, the Riemann Sum will converge to the integral

$$\lim_{n\to\infty} \sum_{i=1}^{n} \frac{(2k-1)\pi}{2n} \sin^2\left(\frac{(2k-1)\pi}{2n}\right) \cdot \frac{\pi}{n} = \int_0^{\pi} x\sin^2(x)\, dx$$

9. (a) Using substitution, evaluate $\displaystyle\int \frac{1}{\sqrt{1+4x}}\, dx$.

Solution: Using substitution, evaluate $\displaystyle\int \frac{1}{\sqrt{1+4x}}\, dx$.

Solution:
Let

$$u = 1 + 4x.$$

Then

$$du = 4\,dx,$$

so that

$$dx = \frac{1}{4}\,du.$$

Substituting,

$$\begin{aligned}
\int \frac{1}{\sqrt{1+4x}}\,dx &= \int \frac{1}{\sqrt{u}} \cdot \frac{1}{4}\,du \\
&= \frac{1}{4}\int u^{-1/2}\,du \\
&= \frac{1}{4}\left(\frac{u^{1/2}}{1/2}\right) + C \\
&= \frac{1}{2}\sqrt{u} + C \\
&= \frac{1}{2}\sqrt{1+4x} + C.
\end{aligned}$$

(b) Evaluate $\int \sec 3x\,dx$.
Solution:
Let

$$u = 3x.$$

Then

$$du = 3\,dx,$$

so that

$$dx = \frac{1}{3}\,du.$$

Substituting,

$$\begin{aligned}
\int \sec(3x)\,dx &= \frac{1}{3}\int \sec(u)\,du \\
&= \frac{1}{3}\ln|\sec(u) + \tan(u)| + C \\
&= \frac{1}{3}\ln|\sec(3x) + \tan(3x)| + C.
\end{aligned}$$

10. (a) Let $f(x) = \frac{1}{2}\left(e^x + e^{-x}\right)$ for $0 \leq x \leq 1$. Determine the arc length of the graph (leave your answer with e in it).

Solution: Solution:

The arc length formula is

$$L = \int_a^b \sqrt{1 + (f'(x))^2}\, dx.$$

First,

$$f'(x) = \frac{1}{2}\left(e^x - e^{-x}\right).$$

So,

$$\begin{aligned} 1 + (f'(x))^2 &= 1 + \left(\frac{1}{2}(e^x - e^{-x})\right)^2 \\ &= 1 + \frac{1}{4}\left(e^{2x} - 2 + e^{-2x}\right) \\ &= \frac{1}{4}\left(e^{2x} + 2 + e^{-2x}\right) \\ &= \left(\frac{1}{2}(e^x + e^{-x})\right)^2. \end{aligned}$$

Thus,

$$\sqrt{1 + (f'(x))^2} = \frac{1}{2}(e^x + e^{-x}).$$

Therefore,

$$\begin{aligned} L &= \int_0^1 \frac{1}{2}(e^x + e^{-x})\, dx \\ &= \frac{1}{2}\left[e^x - e^{-x}\right]_0^1 \\ &= \frac{1}{2}\left(e - \frac{1}{e}\right). \end{aligned}$$

(b) Let $f(x) = \frac{4}{3}x^{3/2}$ for $1 \leq x \leq 2$. Determine the arc length of the graph (leave your answer in its exact form...NO DECIMALS). *Hint: Use part (a)*

Solution: Solution:

First,

$$f'(x) = \frac{4}{3} \cdot \frac{3}{2}x^{1/2} = 2\sqrt{x}.$$

Using the arc length formula,

$$
\begin{aligned}
L &= \int_1^2 \sqrt{1+\left(2\sqrt{x}\right)^2}\,dx \\
&= \int_1^2 \sqrt{1+4x}\,dx.
\end{aligned}
$$

Using the result from part (a) of the previous problem,

$$\int \sqrt{1+4x}\,dx = \frac{1}{6}(1+4x)^{3/2} + C.$$

Therefore,

$$
\begin{aligned}
L &= \left[\frac{1}{6}(1+4x)^{3/2}\right]_1^2 \\
&= \frac{1}{6}\left(9^{3/2} - 5^{3/2}\right) \\
&= \frac{1}{6}\left(27 - 5\sqrt{5}\right).
\end{aligned}
$$